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FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2025
ENVIRONMENTAL ENGINEERING AND SUSTAINABILITY

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART-A			**5 x2=10**
1.	**a.**	How does the production of biogas contribute to waste management and environmental sustainability?	L2	CO1	**02**
	b.	Demonstrate the key advantage of using aluminum in electronic devices and briefly explain how it contributes to sustainability.	L3	CO2	**02**
	c.	Identify the primary function of ultraviolet radiation in water treatment.	L2	CO3	**02**
	d.	Discuss one compelling reason for the need to recycle electronic waste.	L2	CO4	**02**
	e.	State the objectives of the Water (Prevention and Control of Pollution) Act and briefly outline its scope.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Analyze and discuss the economic, environmental, and social merits and demerits of utilizing biomass energy as a renewable resource.	L3	CO1	**05**
		OR			
	b.	Identify and analyze two significant technical challenges associated with the storage of hydrogen for fuel cells.	L2	CO1	**05**
3.	**a.**	Write the innovative applications of graphene in electronic gadgets. Discuss its unique properties that make it a promising green material for electronics.	L2	CO2	**05**
		OR			
	b.	Analyze the challenges associated with solar chargers and propose technological advancements to improve their efficiency and usability. Discuss the potential role of solar chargers in promoting renewable energy adoption.	L3	CO2	**05**
4.	**a.**	Conduct a comparative analysis between activated carbon and reverse osmosis in water purification.	L3	CO3	**05**

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		OR			
	b.	Explain the working principle of electronic water purifiers and discuss two advantages they offer over non-electronic methods.	L2	CO3	**05**
5.	**a.**	Provide a comprehensive overview of the electronic waste recycling process. Discuss the key stages involved and the importance of each stage in the overall process.	L3	CO4	**05**
		OR			
	b.	Elaborate on the pyro-metallurgical processing method used in electronic waste recycling. Discuss its advantages and limitations.	L2	CO4	**05**
6.	**a.**	Identify and analyze the challenges in the effective implementation of Plastic Waste Management Rules. Propose viable solutions to address these challenges	L3	CO5	**05**
		OR			
	b.	Discuss the role of major environmental acts in promoting sustainable development. Evaluate their contributions to balancing environmental conservation and development needs.	L3	CO5	**05**
7.	**a.**	Compare and contrast two different wind energy conversion technologies, highlighting their strengths and limitations.	L3	CO1	**05**
		OR			
	b.	Examine the properties of borosilicate glass and its advantages in electronics. Provide examples of electronic gadgets where borosilicate glass is commonly used.	L2	CO2	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Provide a comprehensive overview of the importance of integrating various renewable energy sources for achieving a sustainable and resilient energy future. Discuss potential challenges and strategies for successful integration.	L3	CO1	**10**
		OR			
	b.	Provide a comprehensive understanding of the working principles of a wind turbine, discussing the key components and mechanisms involved.	L2	CO1	**10**
9.	**a.**	Evaluate the sustainability of iron alloys in electronic devices, considering their environmental impact and longevity. Provide examples of electronic gadgets where iron alloys are commonly used.	L3	CO2	**10**
		OR			

	b.	Discuss on a green charger, considering energy efficiency and the selection of sustainable materials. Discuss how such a charger contributes to eco-friendly charging practices and reduces electronic waste.	L3	CO2	**10**
10.	**a.**	Design a solar-powered water filtration system and discuss its potential applications. Evaluate the advantages and challenges of using solar energy for water purification.	L6	CO3	**10**
		OR			
	b.	Examine the role of water purification technologies, including both non-electronic and electronic methods, in addressing global water challenges. Discuss their significance in achieving sustainable water management goals.	L3	CO3	**10**
11.	**a.**	Explain the concept of Extended Producer Responsibility (EPR) in the context of e-waste management. Discuss its effectiveness in promoting sustainable practices.	L2	CO4	**10**
		OR			
	b.	Identify and analyze the current challenges in e-waste recycling. Provide insights into the future outlook of the industry, considering technological advancements and policy changes.	L3	CO4	**10**
12.	**a.**	Discuss the role of Non-Governmental Organizations (NGOs) in promoting effective waste management practices. Evaluate their contributions and challenges.	L3	CO5	**10**
		OR			
	b.	Assess the impact of major waste management rules on industries. Discuss how these regulations influence industrial practices and sustainability.	L3	CO5	**10**
13.	**a.**	Conduct a comparative analysis between electric and non-electric water purifiers. Discuss their strengths, weaknesses, and situations where one may be preferred over the other.	L3	CO3	**10**
		OR			
	b.	Explore the economic opportunities associated with the e-waste recycling industry. Discuss job creation, resource recovery, and potential economic benefits for stakeholders.	L3	CO4	**10**